Machine Learning and Probability: Bayesian Review

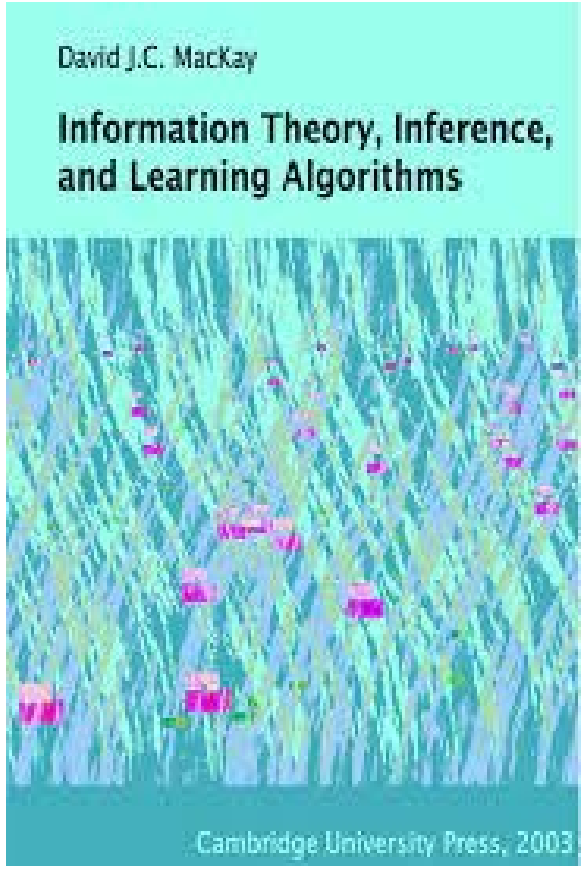
Lecture 3

Tuesday March 10, 2026

NEXT: **Lecture 4**, Friday, March 13: 14:30–16:00 CET
Zoom Link: https://unipd.zoom.us/j/85246961830
Meeting ID: **852 4696 1830**

Suggested Reference: It's free!

https://www.inference.org.uk/itprnn/book.pdf



Bayesian Inference + Information Theory

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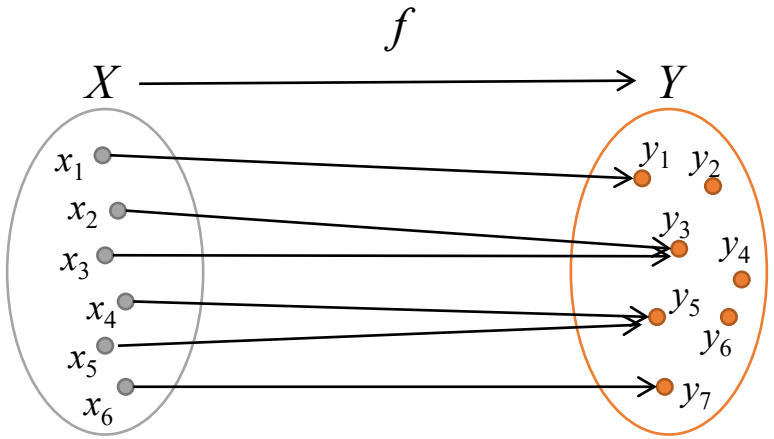
From Functions to Probability

Relaxing deterministic relations into stochastic relations

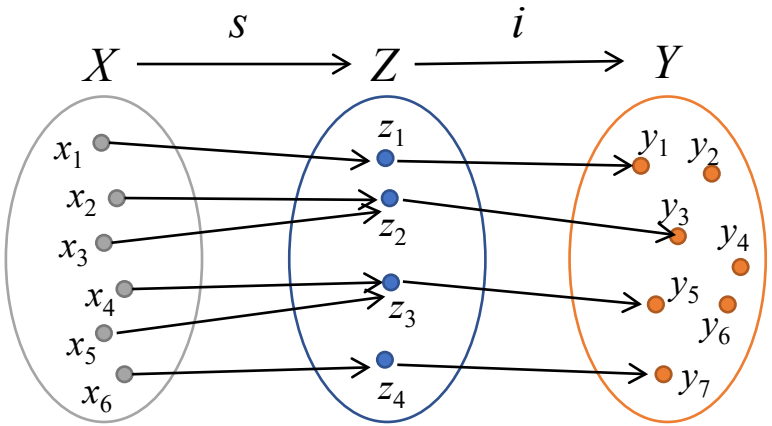
Functions: Measuring information loss

Use the function f to partition the set X into disjoint non-empty subsets. The cells in the partition are equivalence classes containing those $f(x)$ that map to the same value in Y.



Surjective: f assigns an element of X to every element of Y and the pre-image of f for some element in Y has multiple elements in X

Injective: f assigns a unique element of Y to each element of X but not for every element of Y.



https://en.wikipedia.org/wiki/Equivalence_relation#Equivalence_class,_quotient_set,_partition

Subsets of the domain:

"One-to-one" subset $=\{y_1, y_7\}$

"Multi-hit" subset $=\{y_3, y_5\}$

"No hit" subset $=\{y_2, y_4, y_6\}$

Quotient Set: $Z = X/\sim$

Z = the "level" sets of f

$z_1=\{x_1\}\sim y_1$

$z_2=\{x_2,x_3\}\sim y_3$

$z_3=\{x_4,x_5\}\sim y_5$

$z_4=\{x_6\}\sim y_7$

Functions: Measuring information loss

How can a function between two sets X and Y, $f: X \rightarrow Y$, fail to represent all the reconstructive information? In other words, given an element of Y, what is the corresponding element in X?

- Assigning *multiple* elements of set X to the same element of set Y.
- Failing to assign *any* element of set X to an element of set Y.

In this context, the *best* function would be a bijection, mapping each element of X to a *unique* element of Y for every Y. But this requires the cardinality of the sets to be equal.

$$X \xrightarrow{f} Y \quad \Rightarrow \quad \overbrace{X \underset{\text{covers target}}{\xrightarrow{s}} Z \underset{\text{one-to-one}}{\xrightarrow{i}} Y}^{f = i \circ s}$$

$$Z \doteq \{ z \in X \mid y = f(x) \}$$

Every function has a *canonical factorization* into the composition of an injective with a surjective function. The failure of a function to be a bijection can be measured by its factorization.

What is a model? A compressed data representation?

Generative Models can recreate a version of the data as restricted by the relevance:

Lossless	Data Transformations	***Lossy***
data structure	⟷	*model*
$f = i$	$f = i \circ s$	$f = s$

Whether the 'model' is good depends on how well it preserves the relevant information.

The data can be a combination of deterministic (*signal*) and a stochastic (n*oise*) model:

$$data = signal + noise$$

BUT noise is not "something" out in the World, it's a MODEL choice!

Probability: Relaxing the idea of a function

A deterministic *function* is a binary relation between the sets X and Y with the restriction that each element of X is assigned to a unique element of Y:

$$f : X \to Y$$

The *conditional probability*, $p(y|x)$, is *not* a function from X to Y but rather a function from X to a probability simplex, $\Delta(Y)$, on Y:

$$\Delta(Y) = \left\{ p : Y \to [0,1],\ \sum_y p(y) = 1 \right\}$$

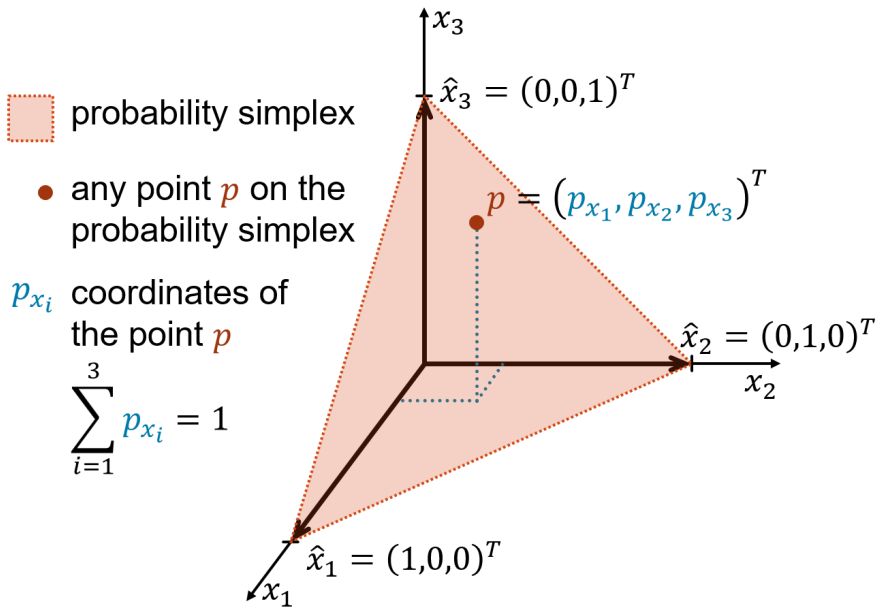
Probabilistic function: $\mathcal{P} : X \to \Delta(Y)$

$$x \mapsto p(y \mid x)$$

Deterministic function is a special case:

$$\mathcal{P}_f : X \to \text{one-hot at } f(x)$$

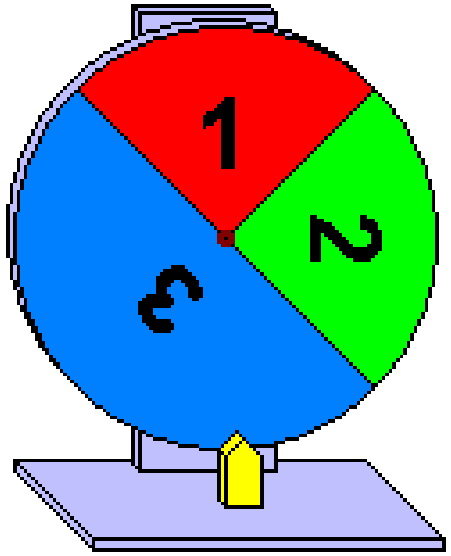
$$x \mapsto p_f(y \mid x) = \delta_{y, f(x)}$$



The Ensemble or Probability Space or "Triple"

An ensemble X is a triple $(x, \mathcal{A}_X, \mathcal{P}_X)$, where the *outcome* x is the value of a random variable, which takes on one of a set of possible values, $\mathcal{A}_X = \{a_1, a_2, \ldots, a_i, \ldots, a_I\}$, having probabilities $\mathcal{P}_X = \{p_1, p_2, \ldots, p_I\}$, with $P(x = a_i) = p_i$, $p_i \geq 0$ and $\sum_{a_i \in \mathcal{A}_X} P(x = a_i) = 1$.

SAMPLE SPACE



$$\mathcal{A}_X = \{1, 2, 3\}$$

Outcome or Elementary Event:
A single execution of the model.

$$x$$

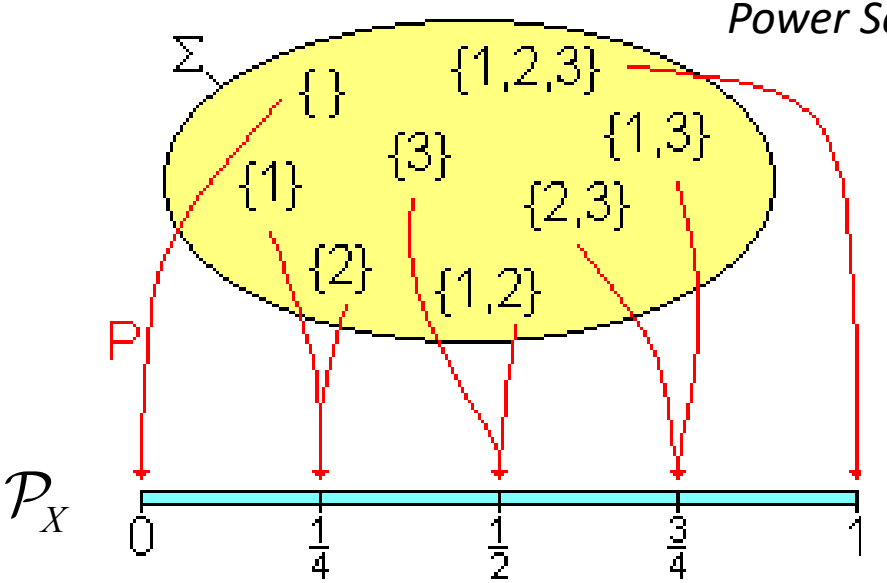
One of the numbers is *randomly* (?) selected by the process.

Is this the **WORLD**?
Or is it a **MODEL** of the World?

Did you observe one of the numbers listed in the event?

EVENT SPACE

Power Set



PROBABILITY MEASURE
Assigns a probability to each event

Probability is in our Models!!!
Avoid the Mind Projection Fallacy!

Probability quantifies uncertainty in our representations and the corresponding inference procedures from data to models. Failure to understand this will cause our minds to get mixed into the physical world in a very confusing way.

Do ***NOT*** physicalize probability.

Probability: The Logic of Uncertainty

Probability is nothing but common sense reduced to calculation. – Pierre-Simon Laplace

MacKay, *ITILA* (2003)

Notation. Let 'the degree of belief in proposition x' be denoted by $B(x)$. The negation of x (NOT-x) is written $\overline{x}$. The degree of belief in a conditional proposition, 'x, assuming proposition y to be true', is represented by $B(x \mid y)$.

Axiom 1. Degrees of belief can be ordered; if $B(x)$ is 'greater' than $B(y)$, and $B(y)$ is 'greater' than $B(z)$, then $B(x)$ is 'greater' than $B(z)$.

[Consequence: beliefs can be mapped onto real numbers.]

Axiom 2. The degree of belief in a proposition x and its negation $\overline{x}$ are related. There is a function f such that

$$B(x) = f[B(\overline{x})].$$

Axiom 3. The degree of belief in a conjunction of propositions x, y (x AND y) is related to the degree of belief in the conditional proposition $x \mid y$ and the degree of belief in the proposition y. There is a function g such that

$$B(x, y) = g\left[B(x \mid y), B(y)\right].$$

Box 2.4. The Cox axioms.
If a set of beliefs satisfy these axioms then they can be mapped onto probabilities satisfying $P(\text{FALSE}) = 0$, $P(\text{TRUE}) = 1$, $0 \leq P(x) \leq 1$, and the rules of probability:

$$P(x) = 1 - P(\overline{x}),$$ **SUM** rule

and

$$P(x, y) = P(x \mid y)P(y).$$ **PRODUCT** rule

Rules of Probability

PRODUCT rule

$$x \in \mathcal{A}_X \rightleftarrows y \in \mathcal{A}_Y$$

$$P(X \mid Y) \cdot P(Y) = P(X, Y) = P(Y \mid X) \cdot P(X)$$

Bayes' Rule:

$$P(X \mid Y) = \frac{P(Y \mid X) \cdot P(X)}{P(Y)}$$

SUM rule

Binary

$$P(x) + P(\sim x) = 1$$

Discrete/*N*-ary

$$\sum_{i=1}^{N} P(x_i) = 1$$

Continuum

$$\int_{x \in \mathcal{A}_X} dx\, p(x) = 1$$

$$y = f(x) \rightarrow x = f^{-1}(y)$$

$$dx = \left|\frac{\partial x}{\partial y}\right| dy = \left|\frac{\partial f^{-1}}{\partial y}\right| dy$$

$$\int_{y \in \mathcal{A}_X} dy \underbrace{\left|\frac{\partial f^{-1}}{\partial y}\right| p\left(f^{-1}(y)\right)}_{\doteq\, p(y)} = 1$$

Marginalization:

$$P(Y) = \sum_X P(Y \mid X) \cdot P(X)$$

Recall from QM! $\langle x | \psi \rangle = \sum_p \langle x | p \rangle \langle p | \psi \rangle \rightarrow \sum_p | p \rangle \langle p | = 1$

Bayesian vs. Frequentist Perspectives

What is on the dashboard of inference?

Coherence: The model's probabilities, posterior updates, and decisions are mutually consistent.

Calibration: The model's confidence and uncertainty estimates match actual performance frequencies.

COMMENTS:

- *Coherence* and *Calibration* are BOTH important goals for statistical inference.
- *Bayesian* work has tended to focus on coherence while *Frequentist* work hasn't been too worried about *Coherence*.
  - The problem with pure *Coherence* is that one can be coherent and completely wrong.
  - Internal coherence of a model does NOT guarantee correspondence to the WORLD!
- *Frequentist* work has tended to focus on *Calibration* while Bayesian work hasn't been too worried about *Calibration*.
  - The problem with pure *Calibration* is that one can be calibrated and completely useless.
  - Calibration must always be stated relative to a target, a population, and a repetition scheme.
- It's important to make use of Bayesian and Frequentist perspectives to achieve a balance.

- Comments from *Michael Jordan*, UC Berkeley

Bayesian Inference + Examples

Bayesian Inference: Looking *backwards* through the model

Physicists usually think about which model can explain the data BUT what about a different view of this question:

How does the data constrain the possible models?

Bayes' Formula:

$$\underbrace{p(\text{MODEL} \mid \overset{\text{Fixed!}}{\text{DATA}})}_{\textbf{POSTERIOR}} = \frac{\overbrace{p(\overset{\text{Fixed!}}{\text{DATA}} \mid \text{MODEL})}^{\textbf{LIKELIHOOD}} \cdot \overbrace{p(\text{MODEL})}^{\textbf{PRIOR}}}{\underbrace{p(\underset{\text{Fixed!}}{\text{DATA}})}_{\textbf{EVIDENCE}}}$$

Assumptions

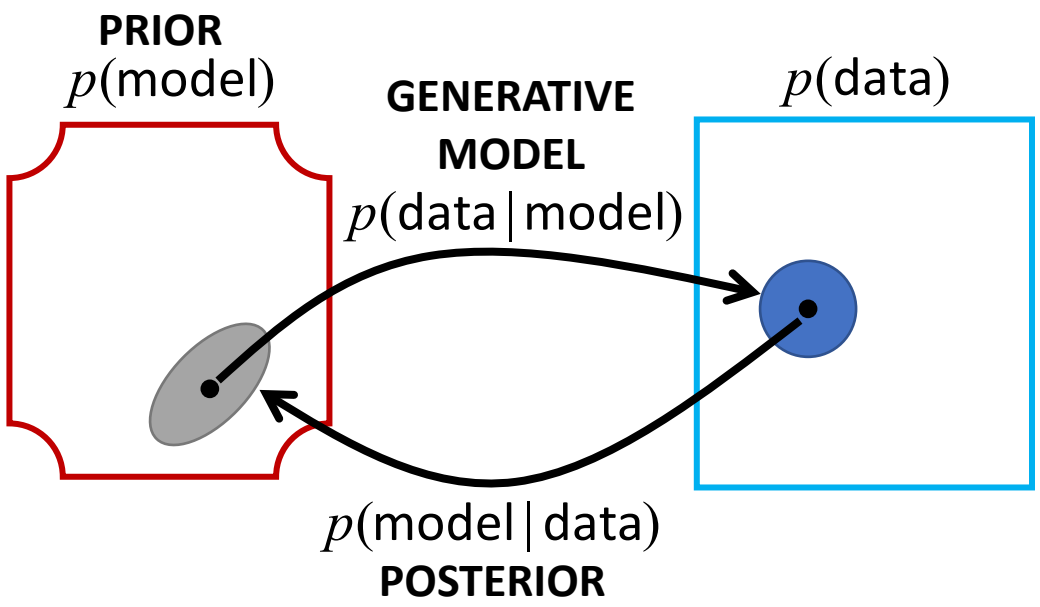
The models are instances of a stochastic process that fluctuates around the data!

Statistical Resolvability of a Model from Data

Sensors should be capable of observing much more than they actually do.
Always recognize this difference between what *could* have been observed and what was *actually* observed.
It is in this difference that models emerge as constraints on what we do observe in the physical world …

- What could be observed? → p(data)
- What is observed? → data = obs.
- What is the resolution of the observation? Measurement error.

Bayesians believe they can specify the size of the "Model" box using a prior distribution.



Frequentists believe they can specify a sampling distribution of test statistics for a given model and then compare different models.

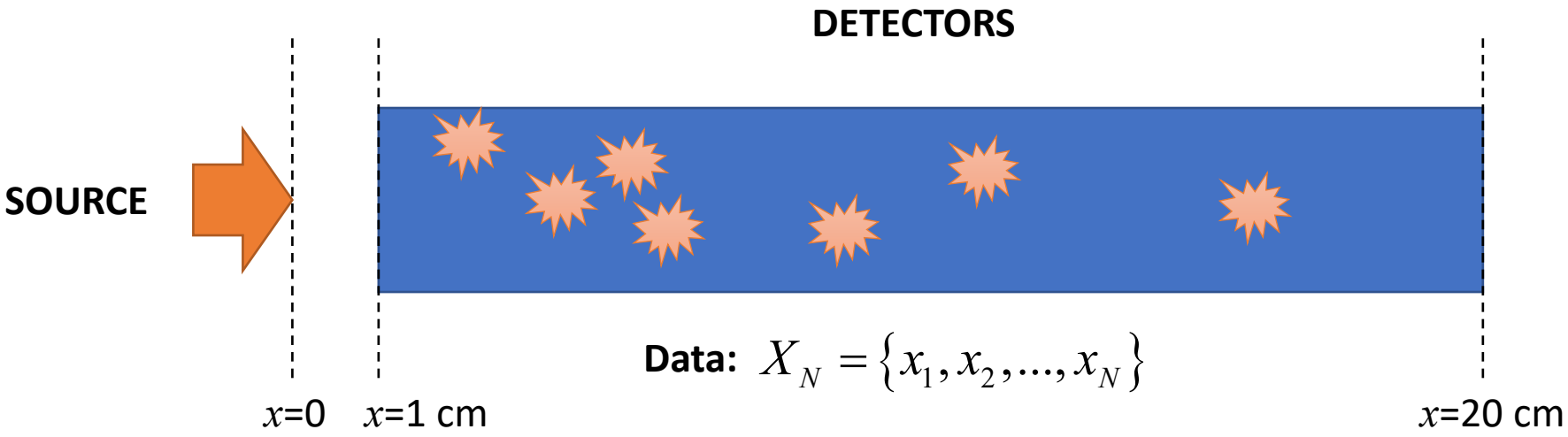
Gull's Decay Length Problem

MacKay, *ITILA*, Chpt. 3, Ex. 3.3, p.47 + detailed solution p.48-50.

Unstable particles are emitted from a source and decay at a distance x, a real number that has an exponential probability distribution with a characteristic length, λ:

$$p(x \mid \lambda) = \frac{e^{-x/\lambda}}{Z}$$

Decay events can *only* be observed if they occur in the window extending from x=1 cm to x=20 cm. N decays are observed at locations $\{x_1, x_2, \ldots, x_N\}$. What is λ?



Gull's Decay Length problem

Prior: $p(\lambda \mid a,b) = \frac{1}{b-a}\mathbf{I}_{[a,b]}$

Or try this one instead: $p(\lambda \mid \alpha,\beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}$

Generative model: $p(x \mid \lambda) = \frac{e^{-x/\lambda}}{Z}, \quad Z(\lambda;a,b) = \int_a^b dx\, e^{-x/\lambda} = \lambda \cdot \left(e^{-a/\lambda} - e^{-b/\lambda}\right)$

Likelihood: N samples, $p\left(X_N \mid \lambda\right) = \prod_{n=1}^{N} \frac{e^{-x_n/\lambda}}{Z(\lambda)} = \frac{\lambda^{-N} e^{-\bar{x}/\lambda}}{\left(e^{-a/\lambda} - e^{-b/\lambda}\right)^N}$

Data: $X_N = \{x_1, x_2, \ldots, x_N\}$

$$\bar{x} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

Evidence: $p\left(X_N\right) = \int_0^{\infty} d\lambda\, p\left(X_N \mid \lambda\right) p(\lambda) = \frac{1}{b-a}\int_a^b d\lambda \frac{\lambda^{-N} e^{-\langle x\rangle/\lambda}}{\left(e^{-a/\lambda} - e^{-b/\lambda}\right)^N}$

Bayes' formula: $p\left(\lambda \mid X_N\right) = \frac{p\left(X_N \mid \lambda\right)\cdot p(\lambda)}{p\left(X_N\right)} \propto \left[\lambda \cdot (e^{-a/\lambda} - e^{-b/\lambda})\right]^{-N} e^{-\langle x\rangle/\lambda} \cdot \mathbf{I}_{[a,b]}(\lambda)$

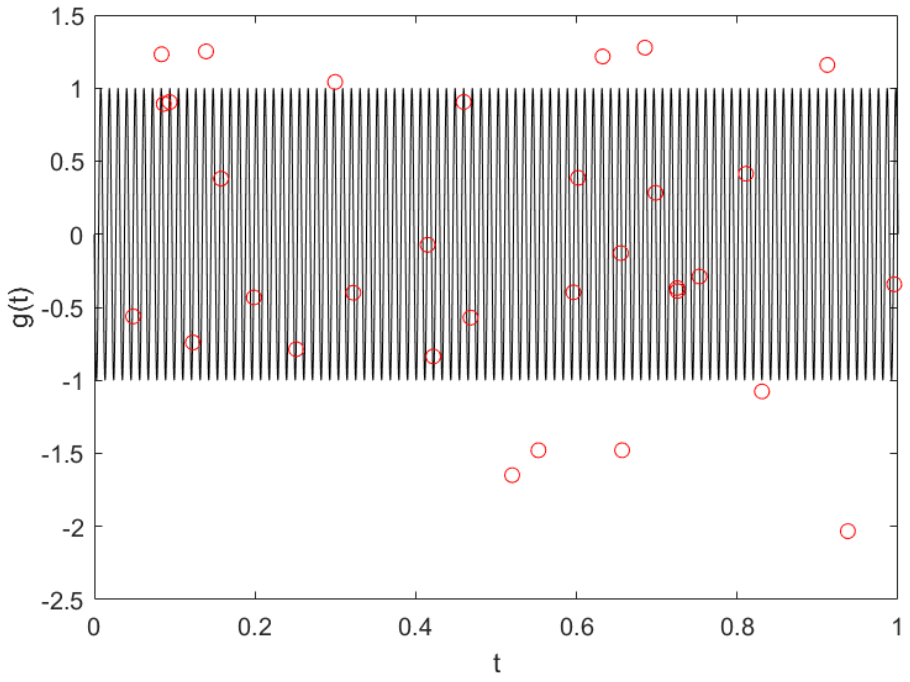
Bayesian Inference of a Sinusoidal Frequency

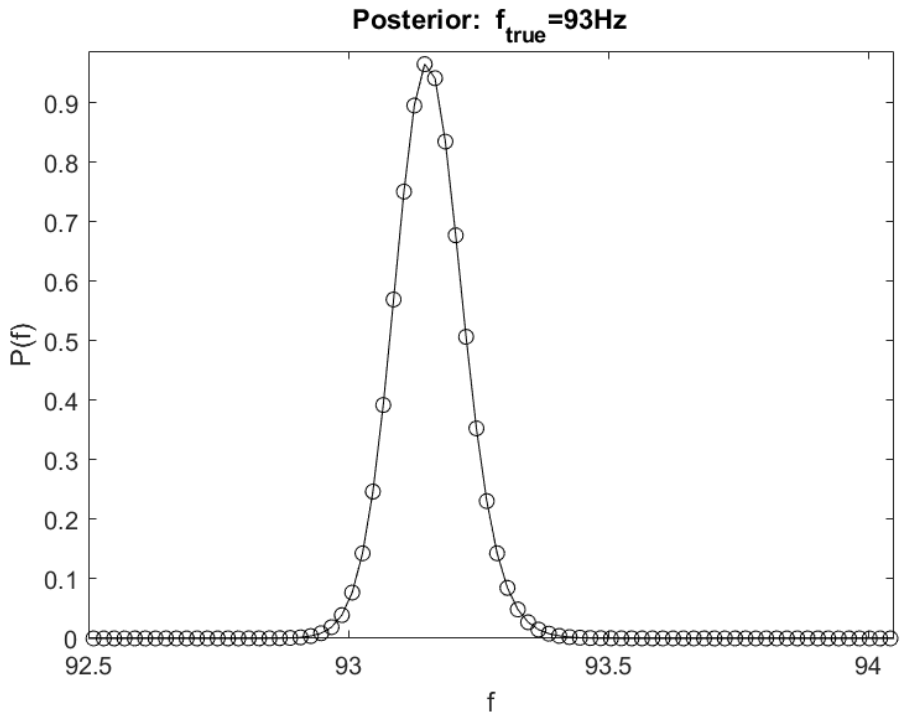
Amazing results if your prior is a good choice!

DATA: $\{(t_n, y_n) \mid n = 1, \ldots, N\}$

MODEL: $y = A_1 \cdot \sin(2\pi ft) + A_2 \cdot \cos(2\pi ft)$

$\rightarrow y = A \cdot \sin(2\pi ft + \phi)$





Infer the frequency, *f*, *and* phase, ϕ, from DATA

Convert to polar coordinates: $dA_1 dA_2 = A dA d\phi$

Generative model: $p\left(y \mid A,\phi,f,\sigma;t\right) = \dfrac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}\left(y - A\sin(2\pi f t+\phi)\right)^2}$

Likelihood: $p\left(D \mid A,\phi,f,\sigma^2;I\right) = \dfrac{\sigma^{-N}}{(2\pi)^{N/2}} e^{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n - A\sin(2\pi f\cdot t_n+\phi)\right)^2}$

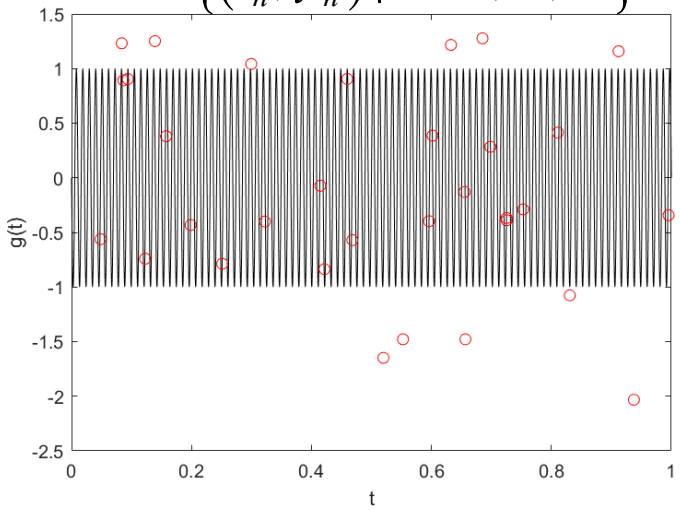
Prior: $p\left(A,\phi,f,\sigma \mid I\right) = \underbrace{p(A \mid I)}_{\frac{1}{A_c}\Theta(A_c - A)} \cdot \underbrace{p(\phi \mid I)}_{\frac{1}{2\pi}} \cdot \underbrace{p(f \mid I)}_{\frac{1}{f_c}\Theta(f_c - f)} \cdot \underbrace{p(\sigma \mid I)}_{\frac{1}{\sigma}}$

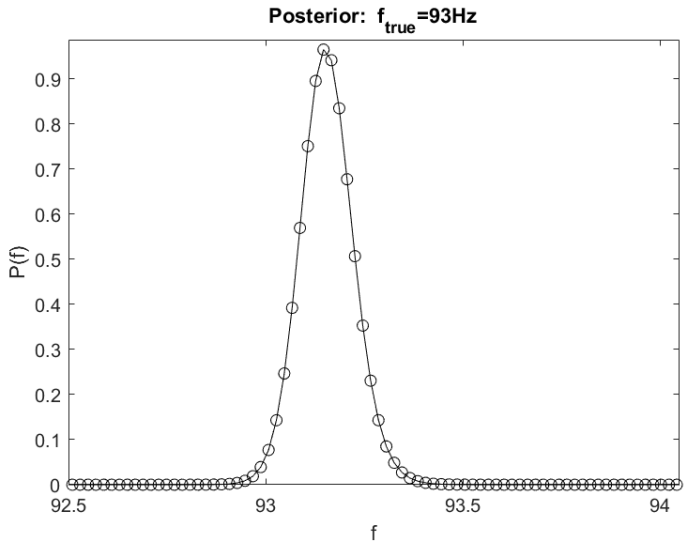
Posterior: $p\left(A,\phi,f,\sigma \mid D,I\right) = \dfrac{p\left(D \mid A,\phi,f,\sigma,I\right)\cdot p\left(A,\phi,f,\sigma \mid I\right)}{p\left(D \mid I\right)}$

Marginal posterior removing A and σ:

$$p(f,\phi \mid D,I) = \int_0^\infty d\sigma \int_0^\infty dA A \; p\left(A,f,\phi,\sigma^2 \mid D,I\right)$$

$$D = \left\{\left(t_n, y_n\right) \mid n = 1,\ldots,N\right\}$$

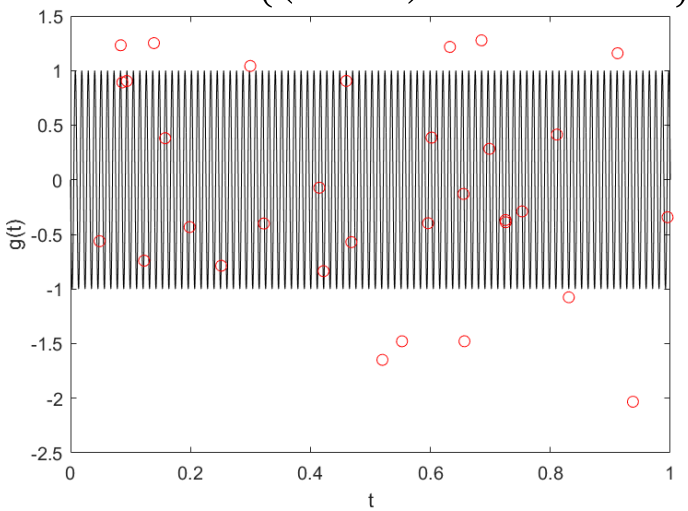




Posterior Probability over Frequency

Lomb-Scargle Periodogram

$$\text{DATA} = \left\{ \left(t_n, y_n \right) \mid n = 1, \ldots, N \right\}$$



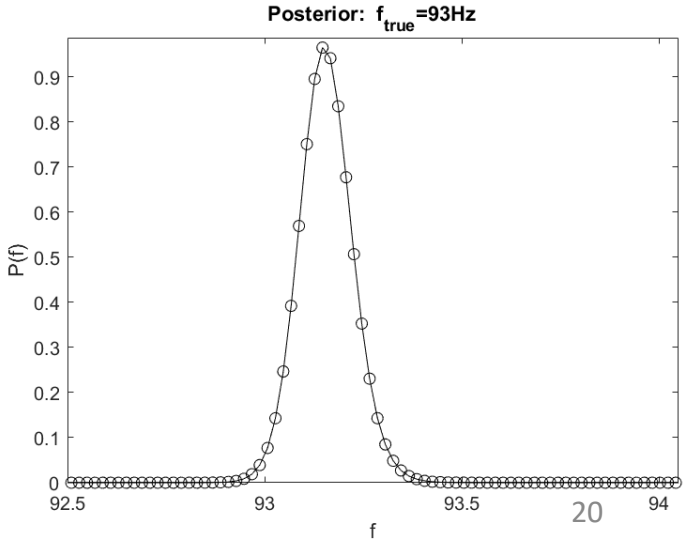
Posterior with amplitudes and noise variance marginalized:

$$p\left(f \mid Y, N, I\right) = \frac{1}{Z}\left(ab - c^2\right)^{-1/2} \cdot \left(1 - \frac{\overline{h^2}}{N\overline{y^2}}\right)^{1-\frac{N}{2}} \qquad \overline{y^2} = \frac{1}{N}\sum_{n=1}^{N} y_n^2$$

$$\overline{h^2} = \frac{bT_1^2 + aT_2^2 - 2cT_1T_2}{ab - c^2}$$

$$T_1 = \sum_{n=1}^{N} y_n \cos\left(2\pi f \cdot t_n\right) \qquad T_2 = \sum_{n=1}^{N} y_n \sin\left(2\pi f \cdot t_n\right)$$

$$a = \sum_{n=1}^{N} \cos^2\left(2\pi f \cdot t_n\right) \qquad b = \sum_{n=1}^{N} \sin^2\left(2\pi f \cdot t_n\right) \qquad c = \sum_{n=1}^{N} \sin\left(2\pi f \cdot t_n\right)\cos\left(2\pi f \cdot t_n\right)$$



Negative log-posterior:

$$-\log p\left(f \mid Y, N, I\right) = \left(N - 2\right) \cdot \frac{1}{2}\log\left(1 - \frac{\overline{h^2}}{N\overline{y^2}}\right) + \frac{1}{2}\log\left(ab - c^2\right) + \log Z$$

NOTE: Need to include the uncertainty when estimating the mean value of the sinusoidal signal. This is especially important for situations with few samples and low SNR.

Infer the frequency, *f*, *and* phase, ϕ, from DATA

Marginal posterior removing A and σ:

$$p(\omega,\phi\,|\,D,I)=\int_0^\infty d\sigma\int_0^\infty dAA\;p\left(A,f,\phi,\sigma^2\,|\,D,I\right)$$

$$\propto\int_0^\infty d\sigma\sigma^{-N-1}\int_0^\infty dAA\;e^{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n-A\sin(\omega\cdot t_n+\phi)\right)^2}$$

#1

$$\sum_{n=1}^{N}\left[y_n-A\sin(\omega t_n+\phi)\right]^2=N\overline{y^2}-2A\underbrace{\sum_{n=1}^{N}y_n\sin(\omega t_n+\phi)}_{T(\omega,\phi)}+A^2\underbrace{\sum_{n=1}^{N}\sin^2(\omega t_n+\phi)}_{\beta(\omega,\phi)}$$

$$=N\overline{y^2}-2AT+A^2\beta=N\overline{y^2}+\left(\beta A^2-2TA+\frac{T^2}{\beta}\right)-\frac{T^2}{\beta}$$

$$=N\overline{y^2}-\frac{T^2}{\beta}+\left(\sqrt{\beta}A-\frac{T}{\sqrt{\beta}}\right)^2$$

$$\propto\int_0^\infty d\sigma\sigma^{-N-1}\int_0^\infty dAA\exp\left\{-\frac{1}{2\sigma^2}\left[N\overline{y^2}-\frac{T^2}{\beta}+\left(\sqrt{\beta}A-\frac{T}{\sqrt{\beta}}\right)^2\right]\right\}$$

$$\propto\int_0^\infty d\sigma\sigma^{-N-1}\exp\left\{-\frac{1}{2\sigma^2}\left(N\overline{y^2}-\frac{T^2}{\beta}\right)\right\}\int_0^\infty dAA\exp\left\{-\frac{1}{2\sigma^2}\left(\sqrt{\beta}A-\frac{T}{\sqrt{\beta}}\right)^2\right\}$$

#2

$$\int_0^\infty dAA\exp\left\{-\frac{1}{2\sigma^2}\left(\sqrt{\beta}A-\frac{T_2}{\sqrt{\beta}}\right)^2\right\}=\beta^{-1}\int_{-\beta^{-1/2}T_2}^{\infty}dx\left(x+\beta^{-1/2}T_2\right)e^{-\frac{x^2}{2\sigma^2}}$$

$$x=\sqrt{\beta}A-\beta^{-1/2}T_2\rightarrow A=\rightarrow dA=\beta^{-1/2}dx$$

$$\propto\beta^{-1}\int_{-\beta^{-1/2}T_2}^{\infty}dx\left(x+\beta^{-1/2}T\right)\int_0^\infty d\sigma\sigma^{-N-1}\exp\left\{-\frac{1}{2\sigma^2}\left(N\overline{y^2}-\frac{T^2}{\beta}+x^2\right)\right\}$$

#3

$$\int_0^\infty d\sigma\sigma^{-(N+1)}e^{-\frac{Q}{2\sigma^2}}=\frac{1}{2}\Gamma\left(\frac{N}{2}\right)\left(\frac{Q}{2}\right)^{-\frac{N}{2}}$$

$$\propto\beta^{-1}\int_{-\beta^{-1/2}T_2}^{\infty}dx\left(x+\beta^{-1/2}T\right)\left(N\overline{y^2}-\beta^{-1}T^2+x^2\right)^{-\frac{N}{2}}$$

Substitution: $x=\beta^{-1/2}T\,z$

$$p(\omega,\phi\,|\,D,I)\propto\beta^{-1}\left(\beta^{-1}T^2\right)^{1-\frac{N}{2}}\int_{-1}^{\infty}dz\left(z+1\right)\left(z^2+N\overline{y^2}\frac{\beta}{T^2}-1\right)^{-\frac{N}{2}}$$

$N\overline{y^2}>\frac{T^2}{\beta}\rightarrow$ No poles on real axis.

$$\overline{h^2}=N\overline{y^2}\cdot\left(\alpha^2+1\right)^{-1}$$

$$\left(\overline{h^2}\right)^{-N/2}=\left(N\overline{y^2}\right)^{-N/2}\cdot\left(\alpha^2+1\right)^{N/2}$$

$$F_N(\alpha^2)=\int_{-1}^{\infty}dz\left(z+1\right)\left(z^2+\alpha^2\right)^{-\frac{N}{2}}\approx\left(\alpha^2\right)^{1-\frac{N}{2}}$$

$$\alpha^2=N\overline{y^2}\frac{\beta}{T^2}-1$$

Sufficient statistic: $\overline{h^2}(\omega,\phi)=\dfrac{T^2(\omega,\phi)}{\beta(\omega,\phi)}=\dfrac{\left(\sum_{n=1}^{N}y_n\sin(\omega t_n+\phi)\right)^2}{\sum_{n=1}^{N}\sin^2(\omega t_n+\phi)}$

Approximately,

$$p(\omega,\phi\,|\,D,I)\propto\frac{1}{\beta(\omega,\phi)}\cdot\left(N\overline{y^2}-\overline{h^2}(\omega,\phi)\right)^{1-\frac{N}{2}}$$

Infer the frequency, *f*, *and* phase, ϕ, from DATA

Relationship to previously used quantities for Lomb-Scargle periodogram

$$\overline{h^2}(\omega,\phi)=\frac{T^2(\omega,\phi)}{\beta(\omega,\phi)}=\frac{\left(\sum_{n=1}^{N} y_n \sin(\omega t_n+\phi)\right)^2}{\sum_{n=1}^{N}\sin^2(\omega t_n+\phi)}$$

$$=\frac{\left(T_1(\omega)\sin\phi+T_2(\omega)\cos\phi\right)^2}{a(\omega)\sin^2\phi+b(\omega)\cos^2\phi+2c(\omega)\sin\phi\cos\phi}$$

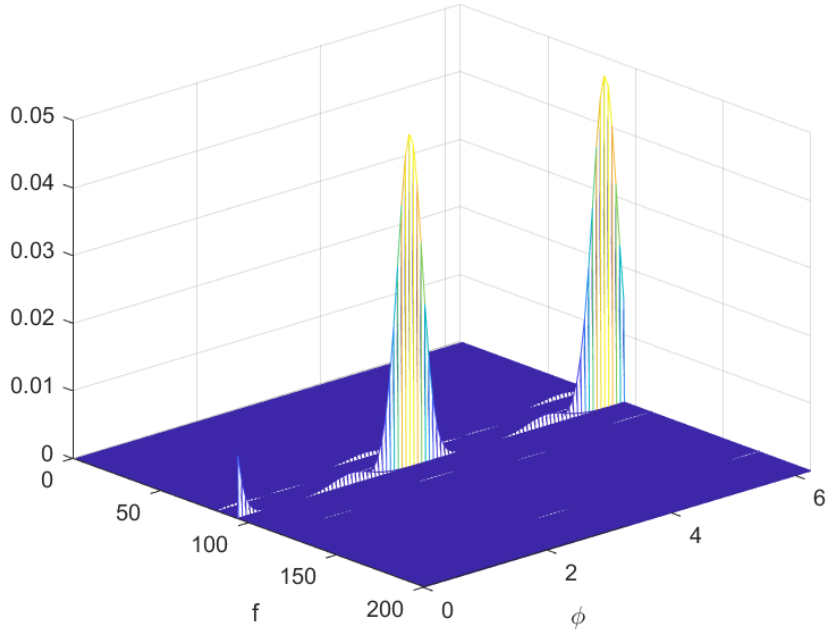
$$T(\omega,\phi)=\sum_{n=1}^{N} y_n \sin(\omega t_n+\phi)=\sin\phi\overbrace{\sum_{n=1}^{N} y_n\cos\omega t_n}^{T_1(\omega)}+\cos\phi\overbrace{\sum_{n=1}^{N} y_n\sin\omega t_n}^{T_2(\omega)}$$

$$\beta(\omega,\phi)=\sum_{n=1}^{N}\sin^2(\omega t_n+\phi)=\sum_{n=1}^{N}\left(\cos\phi\sin\omega t_n+\sin\phi\cos\omega t_n\right)^2$$

$$=\sin^2\phi\underbrace{\sum_{n=1}^{N}\cos^2\omega t_n}_{a(\omega)}+\cos^2\phi\underbrace{\sum_{n=1}^{N}\sin^2\omega t_n}_{b(\omega)}+2\sin\phi\cos\phi\underbrace{\sum_{n=1}^{N}\sin\omega t_n\cos\omega t_n}_{c(\omega)}$$

Approximately,

$$p(\omega,\phi\mid D,I)\propto\frac{1}{\beta(\omega,\phi)}\cdot\left(N\overline{y^2}-\overline{h^2}(\omega,\phi)\right)^{1-\frac{N}{2}}$$



Since,

$$\sin(\omega t_n+\phi\pm\pi)=\overbrace{\cos\pi}^{-1}\cdot\sin(\omega t_n+\phi)+\overbrace{\sin\pi}^{0}\cdot\cos(\omega t_n+\phi)$$
$$=-\sin(\omega t_n+\phi)$$

Symmetry in sufficient statistic:

$$\overline{h^2}(\omega,\phi\pm\pi)=\frac{T^2(\omega,\phi\pm\pi)}{\beta(\omega,\phi\pm\pi)}=\frac{\left(-\sum_{n=1}^{N} y_n\sin(\omega t_n+\phi)\right)^2}{\sum_{n=1}^{N}\left(-\sin(\omega t_n+\phi)\right)^2}=\overline{h^2}(\omega,\phi)$$

Another Question: Is there a sinusoidal signal?

Two hypotheses:

- **H_0**: There is *only* Gaussian noise.
- **H_1**: There is a *single* sinusoidal signal + Gaussian noise

In order compute the probabilities of these different hypotheses, we can no longer marginalize over the amplitudes and variance of the Gaussian noise term.

Instead, to test for a sinusoidal signal, *marginalize* out *frequency* and *phase* to find the probability of amplitude and noise variance.:

$$p\left(A,\phi,f,\sigma^2 \mid D,I\right) = \frac{p\left(D \mid A,\phi,f,\sigma^2,I\right)\cdot p\left(A,\phi,f,\sigma^2 \mid I\right)}{p\left(D \mid I\right)}$$

Sampling distribution: $p\left(y \mid A,\phi,f,\sigma^2;t\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}\left(y - A\sin(2\pi f t+\phi)\right)^2}$

Prior: $p\left(A,\phi,f,\sigma^2 \mid I\right) = \underbrace{p(A \mid I)}_{\frac{1}{A_c}\Theta(A_c - A)} \cdot \underbrace{p(\phi \mid I)}_{\frac{1}{2\pi}} \cdot \underbrace{p(f \mid I)}_{\frac{1}{f_c}\Theta(f_c - f)} \cdot \underbrace{p(\sigma^2 \mid I)}_{\frac{1}{\sigma}}$

Likelihood: $p\left(D \mid A,\phi,f,\sigma^2;I\right) = \frac{\sigma^{-N}}{(2\pi)^{N/2}} e^{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n - A\sin(2\pi f\cdot t_n+\phi)\right)^2}$

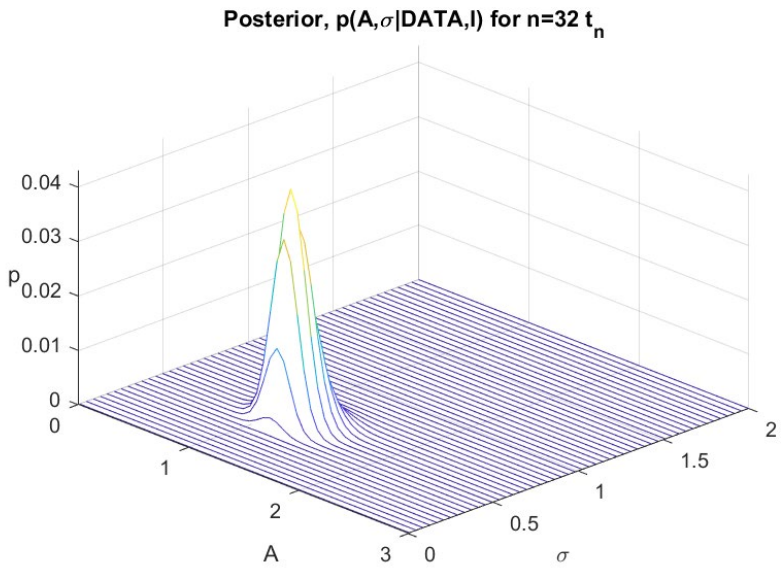
$$p(A,\sigma^2 \mid D,I) = \int_\Omega dA_1 dA_2 df\, \delta\left(A - \sqrt{A_1^2 + A_2^2}\right) p\left(A_1, A_2, f, \sigma^2 \mid D, I\right)$$

Convert to polar coordinates: $dA_1 dA_2 = A dA d\phi$

$$p(A,\sigma^2 \mid D,I) = A \cdot \int_0^\infty df \int_0^{2\pi} d\phi\; p\left(A,\phi,f,\sigma^2 \mid D,I\right)$$

$$p(A,\sigma^2 \mid D,I) = \frac{\Theta(A_c - A)}{2\pi A_c f_c \sigma \cdot p\left(D \mid I\right)} \frac{\sigma^{-N}}{(2\pi)^{N/2}} \int_0^{f_c} df \int_0^{2\pi} d\phi\; e^{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n - A\sin(2\pi f\cdot t_n+\phi)\right)^2}$$

$$= \frac{1}{Z}\sigma^{-N-1} \int_0^{f_c} df \int_0^{2\pi} d\phi\; e^{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n - A\sin(2\pi f\cdot t_n+\phi)\right)^2}$$



From Parameter Estimation to Model Comparison

Parameter estimation for the model $\mathcal{M}_i$:

$$p(\boldsymbol{\theta} \mid \text{data}, \mathcal{M}_i) = \frac{p(\text{data} \mid \boldsymbol{\theta}, \mathcal{M}_i) \cdot p(\boldsymbol{\theta} \mid \mathcal{M}_i)}{p(\text{data} \mid \mathcal{M}_i)}$$

The **evidence** term is not interesting for parameter estimation.

However, the **evidence** term in parameter estimation becomes the **likelihood** for model comparison!

Comparison of the models $\{\mathcal{M}_A, \mathcal{M}_B, \ldots, \mathcal{M}_Z\}$:

$$p(\mathcal{M}_i \mid \text{data}) = \frac{p(\text{data} \mid \mathcal{M}_i) \cdot p(\mathcal{M}_i)}{p(\text{data})}$$

"Marginal Likelihood" OR "Model Evidence"

$$p(\mathcal{M}_i \mid \text{data}) = \frac{\int_{\Omega_i} d\boldsymbol{\theta}\, p(\text{data} \mid \boldsymbol{\theta}, \mathcal{M}_i) \cdot p(\boldsymbol{\theta} \mid \mathcal{M}_i) \cdot p(\mathcal{M}_i)}{\sum_i \int_{\Omega_i} d\boldsymbol{\theta}\, p(\text{data} \mid \boldsymbol{\theta}, \mathcal{M}_i) \cdot p(\boldsymbol{\theta} \mid \mathcal{M}_i) \cdot p(\mathcal{M}_i)}$$

Prior probability of Model

Now, this term is not so interesting!

NOTE: The parameter space, Ω_i, for each model, $\mathcal{M}_i$, can be different.

Bayesian Inference: Looking *backwards* through the model

Goal: *Learn the fairness or bias, B, of a coin.*

- **Assign prior probability based on beliefs:**

  3 different priors (*or initial conditions*)

  BLUE → Coin is heavily biased to **TAILS**

  RED → Coin is biased to **HEADS**

  GREEN → No information (max. entropy)

- **Assign a likelihood to the process:**

$$p(x = \mathrm{H} \mid B) = B$$

$$p(x = \mathrm{T} \mid B) = 1 - B$$

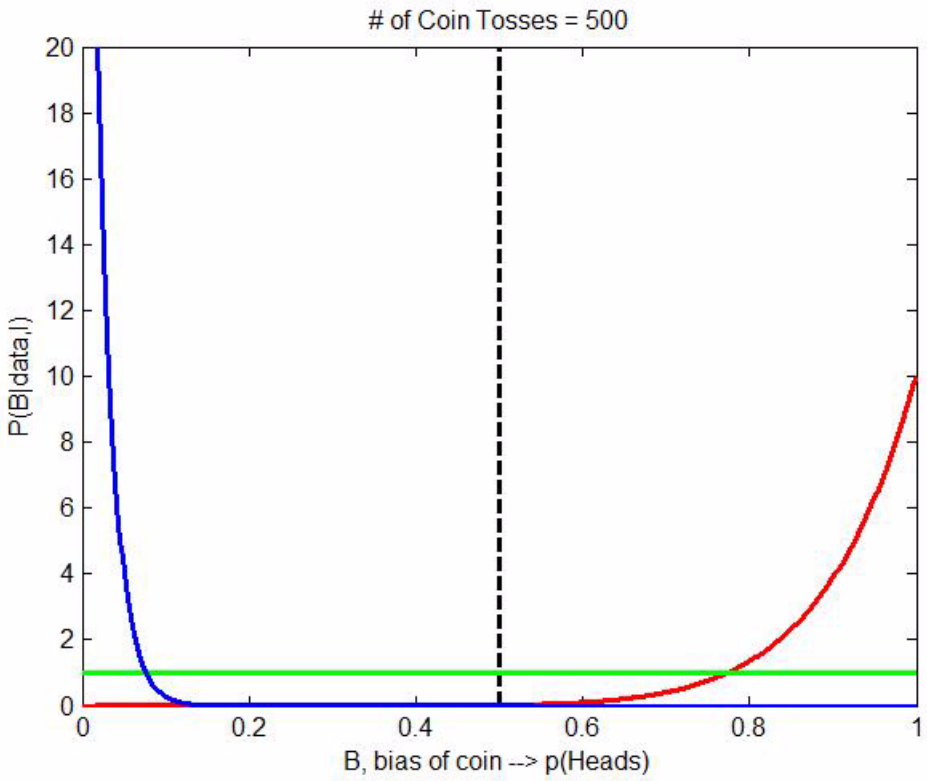
NOTE: *B* is an unknown parameter.

- **Update beliefs with data sequence:**

$$\underbrace{p(B \mid \mathbf{x})}_{posterior} = \frac{\overbrace{p(\mathbf{x} \mid B)}^{likelihood} \cdot \overbrace{p(B)}^{prior}}{\underbrace{p(\mathbf{x})}_{evidence}}$$

Sequence of coin tosses form the data set:

$$\mathbf{x} = \{\mathrm{H\ H\ T\ H\ T\ T\ H\ T\ T\ \ldots\ H\ T\ H}\}$$



Solving the Coin Tossing with a conjugate prior

Likelihood: $$p(N_H \mid B, N) = \binom{N}{N_H} B^{N_H} (1-B)^{N-N_H}$$

Binomial distribution:

$$p(x \mid r, n) = \binom{n}{r} x^r (1-x)^{n-r}$$

$$x \in [0,1] \text{ and } n \geq r \geq 0.$$

Prior: $$p(B \mid \alpha, \beta) = \frac{B^{\alpha-1}(1-B)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}$$

Beta function:

$$\mathrm{B}(\alpha,\beta) = \int_0^1 dx\, x^{\alpha-1}(1-x)^{\beta-1}$$

$$\text{for} \quad \alpha > 0,\ \beta > 0.$$

Jeffrey's prior for Beta distribution: $p(B \mid \frac{1}{2}, \frac{1}{2})$

Uniform prior: $p(B \mid 1,1) = 1$

Evidence: $$p(N_H \mid N; I) = \binom{N}{N_H} \frac{\overbrace{\int_0^1 dB\ B^{N_H+\alpha-1}(1-B)^{N-N_H+\beta-1}}^{\mathrm{B}(N_H+\alpha,\, N-N_H+\beta)}}{\mathrm{B}(\alpha,\beta)}$$

Posterior: $$p(B \mid N_H, N; I) = \frac{B^{N_H+\alpha-1}(1-B)^{N-N_H+\beta-1}}{\mathrm{B}(N_H+\alpha, N-N_H+\beta)}$$

Hyper-parameter update:

$$\alpha \to \alpha + N_H$$

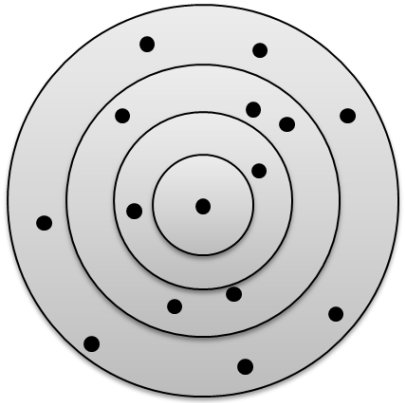
$$\beta \to \beta + N - N_H$$

Model Comparison: Anomaly Detection

Problem Set-up: We are given two datasets, $\mathbf{A}$ and $\mathbf{B}$, and asked if they have been generated by the same process *or* different processes. To complicate matters, there are N_A samples from $\mathbf{A}$ and N_B samples from $\mathbf{B}$..

$\mathbf{A}$ = *Possibly* Anomalous Data $\quad n_A(R)$

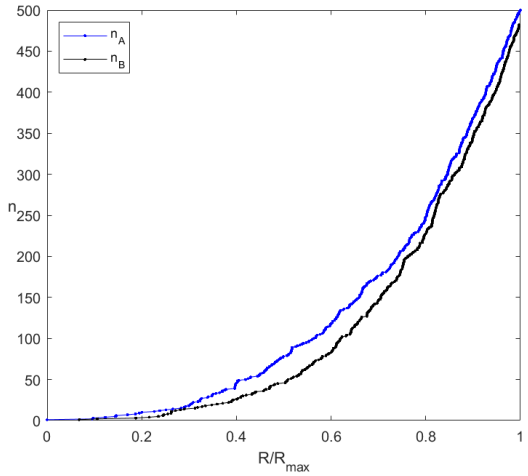
$\mathbf{B}$ = Background Data $\quad n_B(R)$



Approach:

1. Pick a random point in the Background Data, $\mathbf{B}$.
2. Expand a ball of increasing radius, R.
3. Store sample counts, $n_A(R)$ and $n_B(R)$, for both datasets, $\mathbf{A}$ and $\mathbf{B}$.
4. Compare two different models of the probability density:
   - H_0, the probability, b, of samples being in the ball is the same for both data sets.
   - H_1, the probabilities, b_{A} and b_{B}, of samples being in the ball is different for each dataset
5. Select the radius of the ball, R^*, where the probability of H_1 is largest.

→ Repeat process until all data from the $\mathbf{A}$ dataset is "covered".



Model Comparison: Anomaly Detection

$$\mathrm{DATA}_{\mathbf{p}} = \{[n_A(R), N_A], [n_B(R), N_B]\}$$

$$p(H_i \mid \mathrm{DATA}_{\mathbf{p}}) = \frac{p(\mathrm{DATA}_{\mathbf{p}} \mid H_i) \cdot p(H_i)}{p(\mathrm{DATA}_{\mathbf{p}})}$$

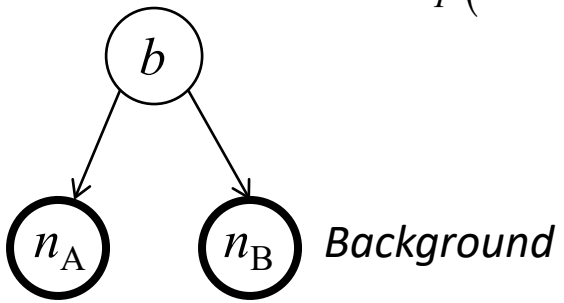
$$p(b \mid \alpha, \beta) = \frac{b^{\alpha-1}(1-b)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}$$

$$\mathrm{B}(\alpha,\beta) = \int_0^1 db\; b^{\alpha-1}(1-b)^{\beta-1}$$

$$p(n \mid b; N) = \binom{N}{n} b^n (1-b)^{N-n}$$

Standard Model

H_0:

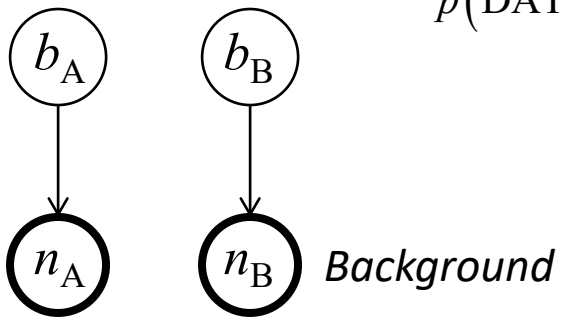


$$p(\mathrm{DATA}_{\mathbf{p}} \mid b; H_0) = p(n_A \mid b; N_A) \cdot p(n_B \mid b; N_B)$$

$$\begin{aligned}
p(\mathrm{DATA}_{\mathbf{p}} \mid H_0) &= \int_0^1 db\; p(\mathrm{DATA}_{\mathbf{p}} \mid b; H_0) \cdot p(b \mid H_0) \\
&= \int_0^1 db\; p(n_A \mid b; N_A) \cdot p(n_B \mid b; N_B) \cdot p(b \mid \alpha, \beta) \\
&= \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \frac{1}{\mathrm{B}(\alpha,\beta)} \int_0^1 db\; b^{n_A+n_B+\alpha-1}(1-b)^{N_A-n_A+N_B-n_B+\beta-1} \\
&= \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \cdot \frac{\mathrm{B}(n_A+n_B+\alpha, N_A-n_A+N_B-n_B+\beta)}{\mathrm{B}(\alpha,\beta)}
\end{aligned}$$

"New Physics" Model

H_1:



$$p(\mathrm{DATA}_{\mathbf{p}} \mid b_A, b_B; H_1) = p(n_A \mid b_A; N_A) \cdot p(n_B \mid b_B; N_B)$$

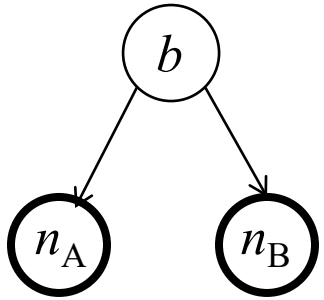
$$\begin{aligned}
p(\mathrm{DATA}_{\mathbf{p}} \mid H_1) &= \int_0^1 db_A \int_0^1 db_B\; p(n_A \mid b_A; N_A) \cdot p(n_B \mid b_B; N_B) \cdot p(b_A \mid \alpha, \beta) \cdot p(b_B \mid \alpha, \beta) \\
&= \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \frac{1}{[\mathrm{B}(\alpha,\beta)]^2} \int_0^1 db_A\; b_A^{n_A+\alpha-1}(1-b_A)^{N_A-n_A+\beta-1} \cdot \int_0^1 db_B\; b_B^{n_B+\alpha-1}(1-b_B)^{N_B-n_B+\beta-1} \\
&= \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \cdot \frac{\mathrm{B}(n_A+\alpha, N_A-n_A+\beta) \cdot \mathrm{B}(n_B+\alpha, N_B-n_B+\beta)}{[\mathrm{B}(\alpha,\beta)]^2}
\end{aligned}$$

Model Comparison: Anomaly Detection

$$\mathrm{DATA}_{\mathbf{p}} = \{[n_A(R), N_A], [n_B(R), N_B]\}$$

$$p(H_i \mid \mathrm{DATA}_{\mathbf{p}}) = \frac{p(\mathrm{DATA}_{\mathbf{p}} \mid H_i) \cdot p(H_i)}{p(\mathrm{DATA}_{\mathbf{p}})} \longrightarrow \log \frac{p(H_1 \mid \mathrm{DATA}_{\mathbf{p}})}{p(H_0 \mid \mathrm{DATA}_{\mathbf{p}})} = \log \frac{p(\mathrm{DATA}_{\mathbf{p}} \mid H_1) \cdot p(H_1)}{p(\mathrm{DATA}_{\mathbf{p}} \mid H_0) \cdot p(H_0)}$$

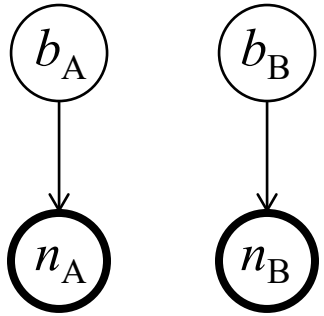
H_0:



$$p(H_0) = \frac{1}{2}$$

$$p(\mathrm{DATA}_{\mathbf{p}} \mid H_0) = \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \cdot \frac{\mathrm{B}(n_A + n_B + \alpha, N_A - n_A + N_B - n_B + \beta) \cdot \mathrm{B}(\alpha, \beta)}{[\mathrm{B}(\alpha, \beta)]^2}$$

H_1:



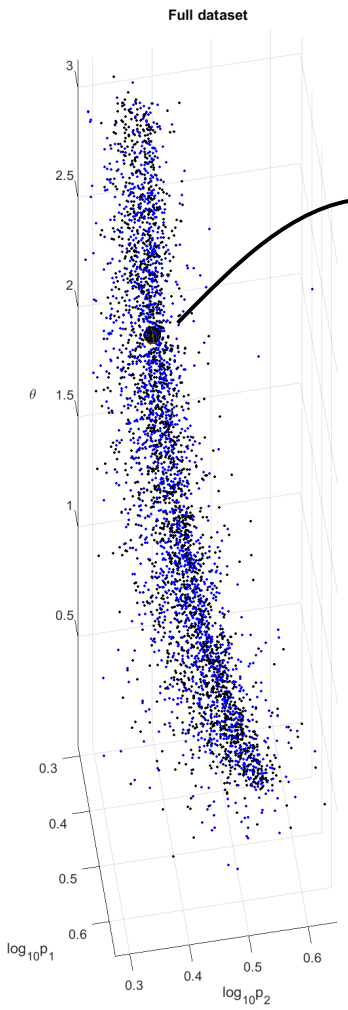
$$p(H_1) = \frac{1}{2}$$

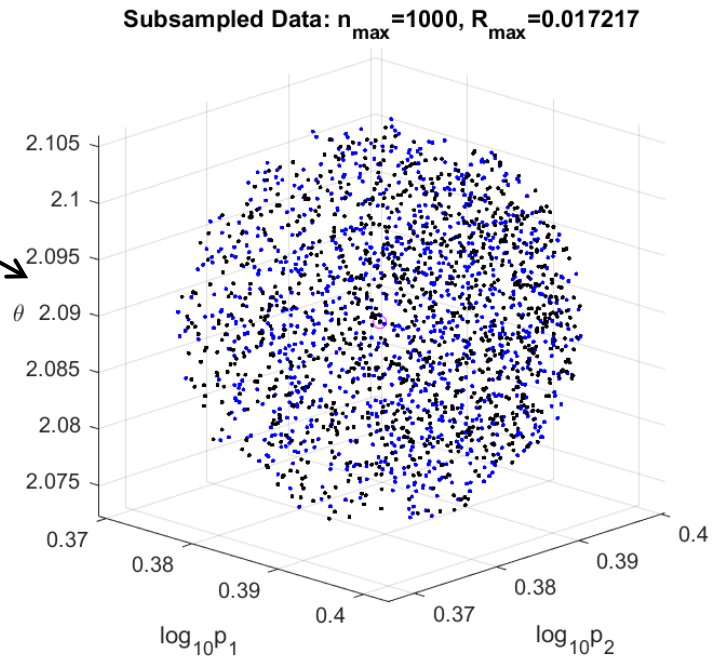
$$p(\mathrm{DATA}_{\mathbf{p}} \mid H_1) = \binom{N_A}{n_A} \cdot \binom{N_B}{n_B} \cdot \frac{\mathrm{B}(n_A + \alpha, N_A - n_A + \beta) \cdot \mathrm{B}(n_B + \alpha, N_B - n_B + \beta)}{[\mathrm{B}(\alpha, \beta)]^2}$$

$\therefore$

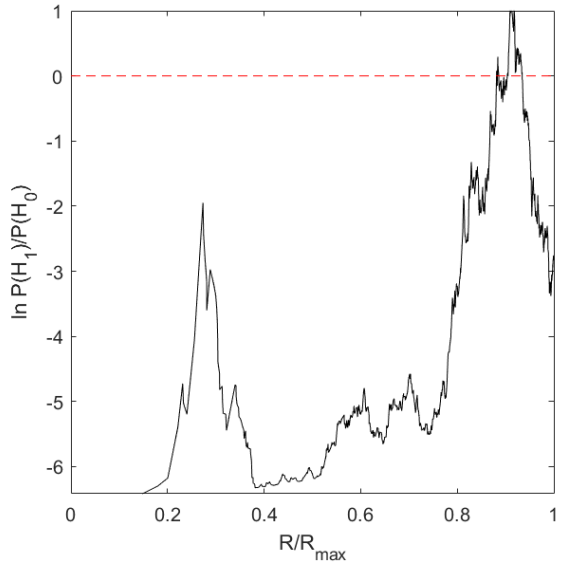
$$\boxed{\log \frac{p(H_1 \mid \mathrm{DATA}_{\mathbf{p}})}{p(H_0 \mid \mathrm{DATA}_{\mathbf{p}})} = \log \frac{\mathrm{B}(n_A + \alpha, N_A - n_A + \beta) \cdot \mathrm{B}(n_B + \alpha, N_B - n_B + \beta)}{\mathrm{B}(n_A + n_B + \alpha, N_A - n_A + N_B - n_B + \beta) \cdot \mathrm{B}(\alpha, \beta)}}$$

Local Anomaly Detection



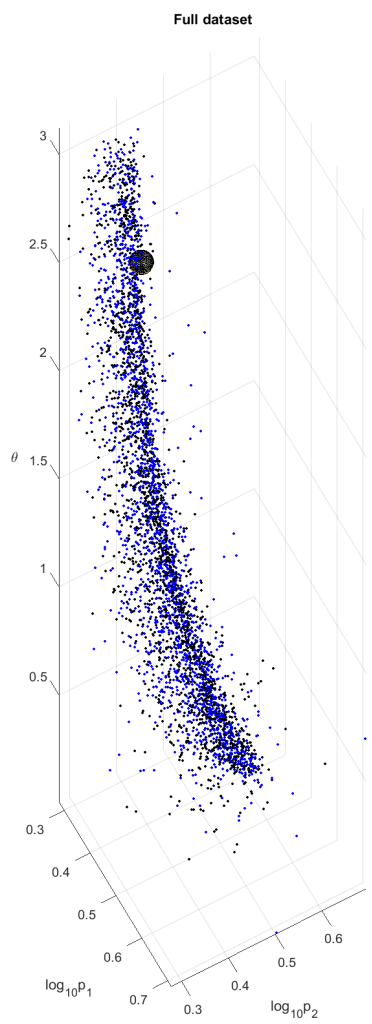


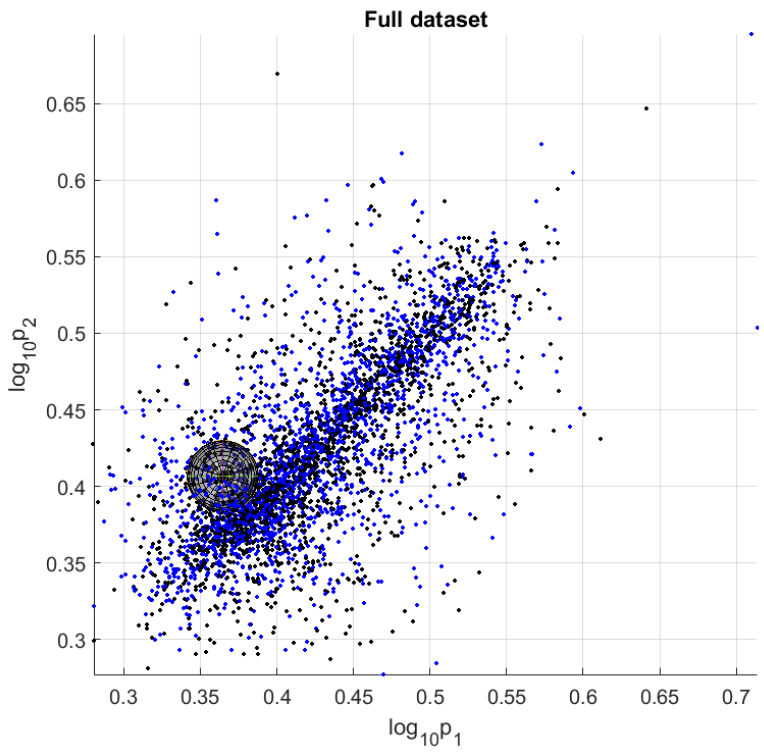
$$t = \log \frac{p(H_1 \mid R)}{p(H_0 \mid R)}$$

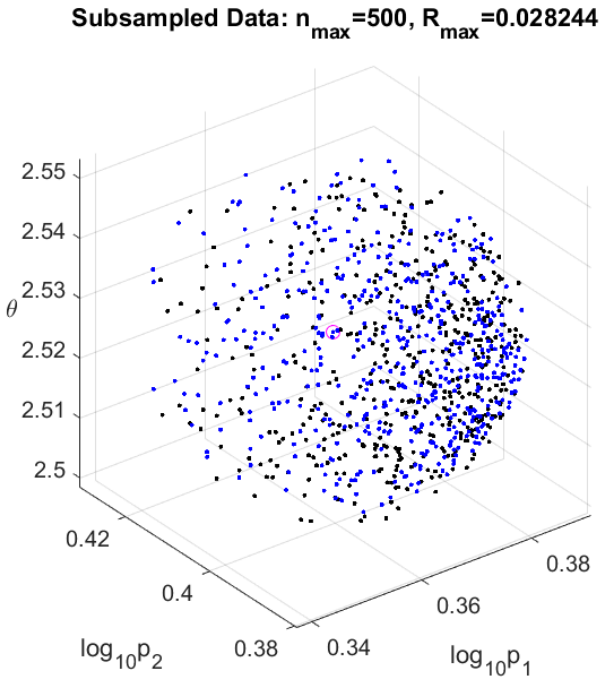


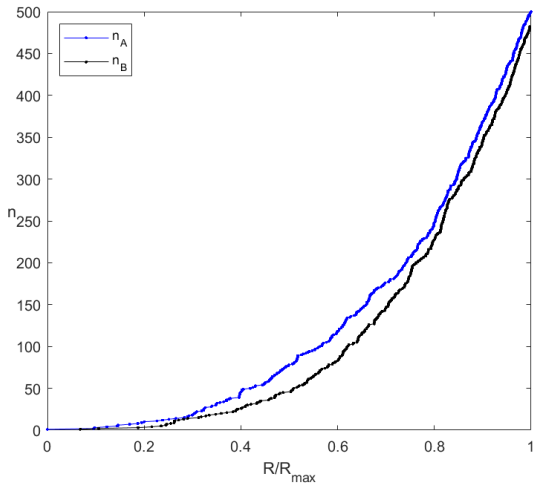
Comparing the Background to Itself

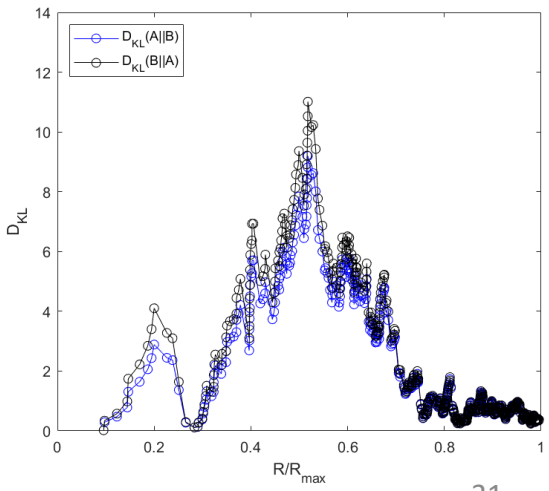
p = [0.3647 0.4066 2.5258]



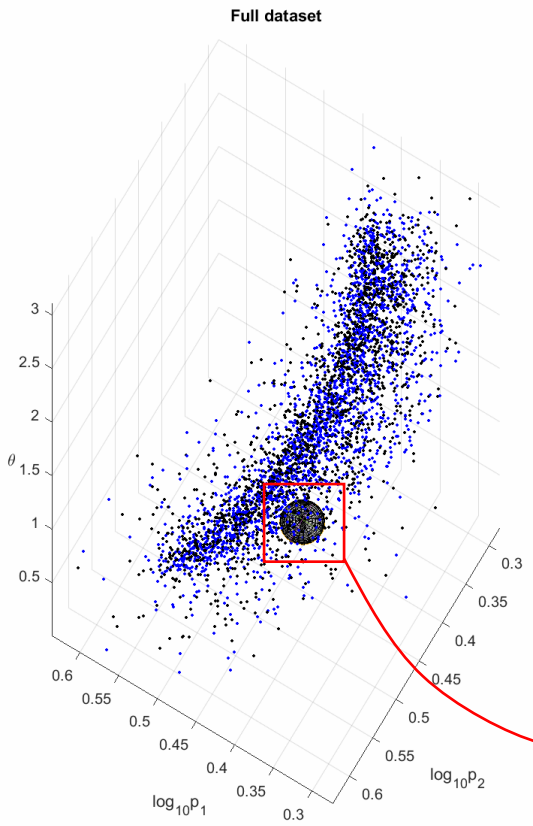


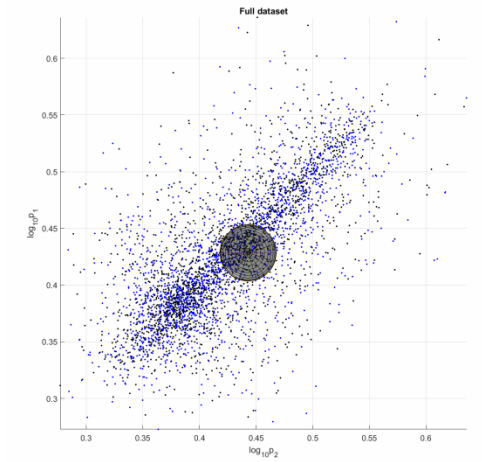


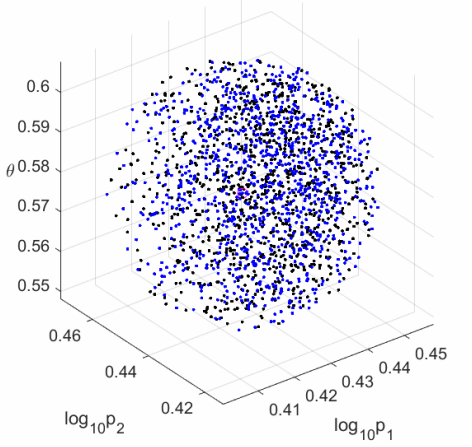




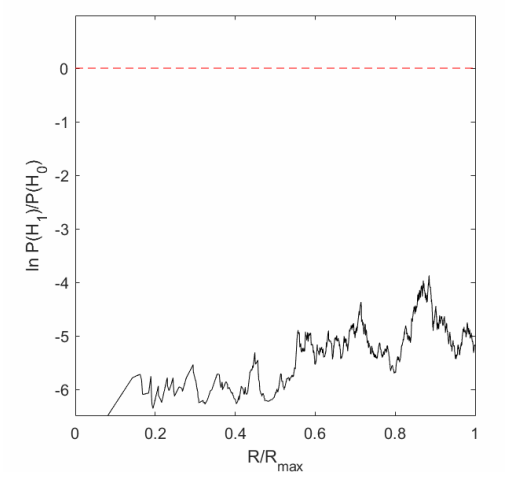
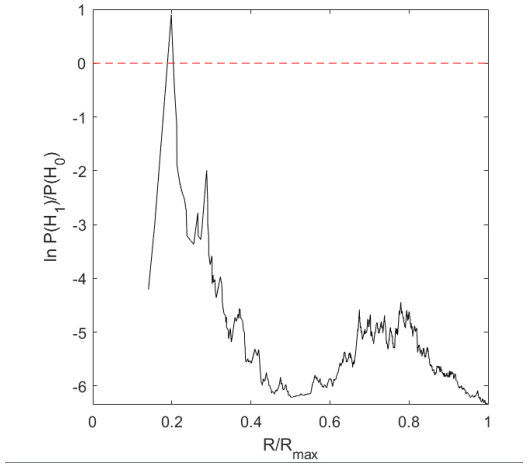
Model Comparison: Anomaly Detection







In control



Out of control